

CM1605 Web Technology

Introduction to HTML II

Week 2 | Janani Harischandra

Learning Outcomes

- Covers LO2 for Module
- On completion of this lecture, students are expected to be able to:
 - Understand and describe HTML Tables, Images, Image maps, HTML Graphics , Colors, Audio/Video Tags, HTML entities.
 - Design hypertext documents using HTML Tables, Images, Image Maps, HTML Audio/Video Tags.

Content

- HTML Tables
- Colors
- Images
- Image Maps
- Audio/Video
- HTML entities and special characters
- HTML Graphics
- HTML5 Tags
- Troubleshooting

HTML Tables

- A table represents information in a grid format.
- There are several types of information that need to be displayed in a grid or table.
- Eg: sports results, stock reports, train time tables, financial reports etc.

HTML Table Example


REUTERS

[World](#)
[Business](#)
[Markets](#)
[Breakingviews](#)
[Video](#)
[More](#)




Commodities

Data delayed by at least 15 min

QUICK MARKET LINKS

- [Stocks](#)
- [Rates & Bonds](#)
- [Currencies](#)

KEY BENCHMARKS

Index	Last	Change	Change %
Baltic Dry Index .BADI	1,598.00	+3.00	+0.19%
4:30 AM +0530			
S&P GSCI TR Index .SPGSCITR	1,784.29	-2.67	-0.15%
9:32 AM +0530			
TR CC/CRB Index .TRCCRB	150.87	+1.66	+1.12%
2:10 AM +0530			
TR Continuous Commodity Index .TRCCI	406.28	+4.60	+1.15%
2:30 AM +0530			

COMMODITIES FUTURES

MARKETS

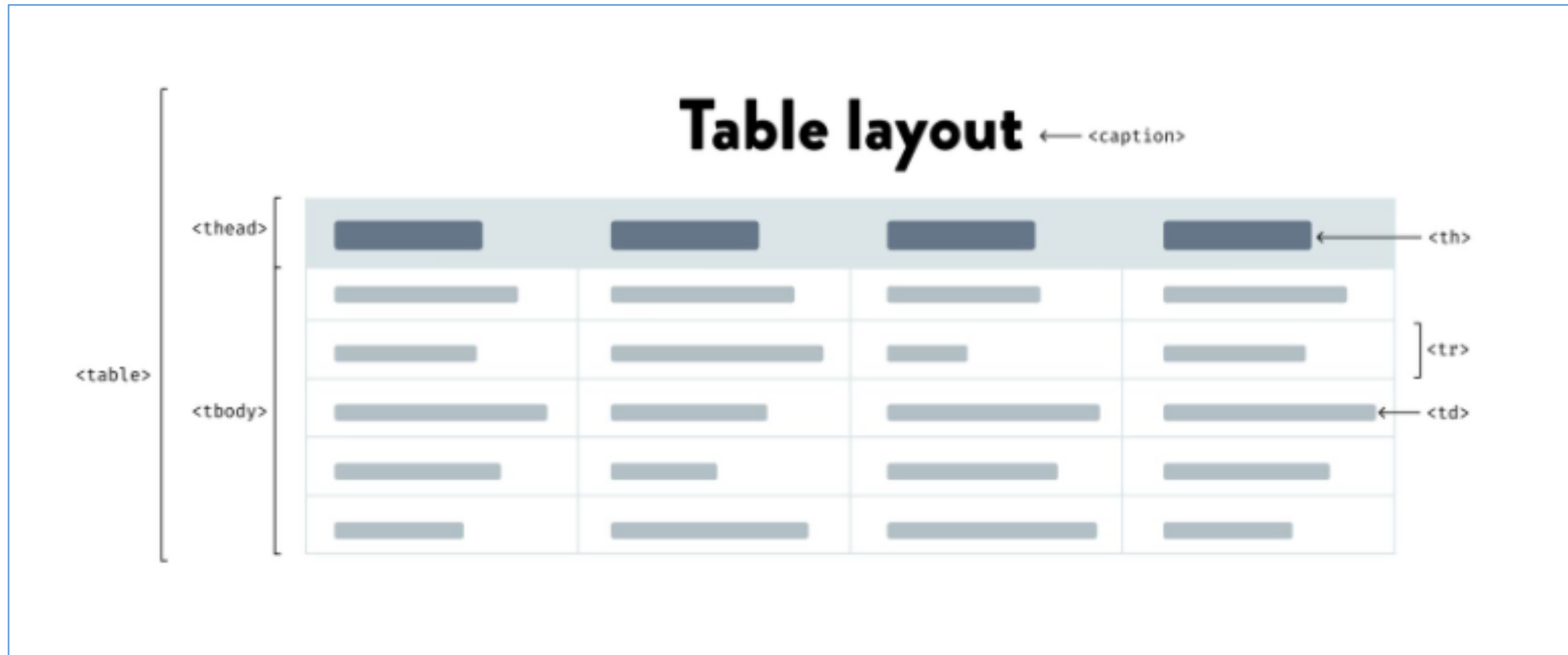
[U.S.](#) [Europe](#) [Asia](#) [MEA](#)

Index	Last	Change
Dow Jones .DJI	27,844.91	-86.11
2:43 AM -0.31%		
+0530		
S&P 500 .SPX	3,381.99	+9.14
2:43 AM +0.27%		
+0530		
Nasdaq .IXIC	11,129.73	+110.42
2:45 AM +1.00%		
+0530		
S&P/TSX .GSPTSE	16,656.12	+141.51
2:35 AM +0.86%		
+0530		

Basic HTML Table Structure

- `<table>`
 - is used to create a table. The contents of the table are written out row by row.
- `<tr>`
 - The start of each row using the opening `<tr>` tag.
- `<td>`
 - Each cell of a table is represented using a `<td>` element.
 - The `<td>` elements are the data containers of the table.
 - They can contain all sorts of HTML elements; text, images, lists, other tables, etc.
- `<th>`
 - Represent the heading for either a column or a row.
 - The text in `<th>` elements are bold and centered.

HTML Table Layout



HTML Tables Attributes

- `colspan` - To make a cell span more than one column
- `rowspan` - To make a cell span more than one row
- `<table border="2">` - To add a border to a table
- `<table cellpadding="2" cellspacing="3">` - Specifies the space between the cell content and its borders
- `<td align="left" align="top">` - To align the content

Table Example

```
<!DOCTYPE html>
<html>
<body>
<h2>Basic HTML Table</h2>
<table border="2">
  <tr>
    <th>Firstname</th>
    <th>Lastname</th>
    <th>Age</th>
  </tr>
  <tr>
    <td>Jill</td>
    <td>Smith</td>
    <td>50</td>
  </tr>
  <tr>
    <td>Eve</td>
    <td>Jackson</td>
    <td>94</td>
  </tr>
</table>
</body>
</html>
```

Basic HTML Table

Firstname	Lastname	Age
Jill	Smith	50
Eve	Jackson	94
John	Doe	80

Exercise – HTML Code?

Quick	brown fox		jumps
over the	lazy	dog	and
	then	it	fall
	prey to a lion		

HTML Colors

- HTML colors can be specified with predefined,
 - Color names - **<h1 style="background-color:Orange;">Orange</h1>**
 - RGB - **rgb(255, 99, 71)**
 - HEX - **#ff6347**
 - HSL - **hsl(9, 100%, 64%)**
 - RGBA - **rgba(255, 99, 71, 0.5)**
 - HSLA - **hsla(9, 100%, 64%, 0.5)**

HTML Colors

Criteria	HTML Markup	Formatted Text
Background Color	<code><h1 style="background-color:DodgerBlue;">Hello World</h1></code>	Hello World
Text Color	<code><h1 style="color:Tomato;">Hello World</h1></code>	Hello World
Border Color	<code><h1 style="border:2px solid Tomato;">Hello World</h1></code>	Hello World

HTML Images

- The HTML tag is used to embed an image in a web page.
- The tag creates a holding space for the referenced image.
- The tag has two required attributes:
 - src - Specifies the path to the image
 - alt - Specifies an alternate text for the image
- The HTML Syntax

```

```

HTML Image Attributes

Attribute	Meaning
alt	- The required alt attribute provides an alternate text for an image, if the user for some reason cannot view it (because of slow connection, an error in the src attribute, or if the user uses a screen reader). This attribute is required in html5.
src	The image URL. This attribute is mandatory
height	Set the height of the image to
width	Set the width of the image

HTML Markup	Formatted Text
<pre></pre>	

HTML Images - Rules

- It is good practice to create a folder for all of the images a website site uses.
- If you have your images in a sub-folder, you must include the folder name in the src attribute:

```

```

- Some web sites store their images on another server.

```

```

- There are three rules to remember when you are adding images for your website
 1. Image format (Jpeg, GIF, PNG)
 2. Image size
 3. Image location

HTML <figure> Tag

- A <figure> element can be used to mark up a photo in a document, and a <figcaption> element to define a caption for the photo.

HTML Markup	Formatted Text
<pre><!DOCTYPE html> <html> <body> <h1>The figure and figcaption element</h1> <figure> <figcaption>Fig.1 - Trulli, Puglia, Italy.</figcaption> </figure> </body> </html></pre>	The figure and figcaption element  Fig.1 - Trulli, Puglia, Italy.

HTML Image Maps

- The HTML <map> tag defines an image map. An image map is an image with clickable areas.
- These clickable areas are defined with one or more <area> tags.
- The image is displayed using the usual tag. Eg: [Click Here](#).

HTML Markup

```
<!DOCTYPE html>
<html>
  <body>
    <h2>Image Maps</h2>
    <p>Click on the computer or the phone to go to a new page and read more about the topic:</p>
    
    <map name="workmap">
      <area shape="rect" coords="34,44,270,350" alt="Computer" href="computer.htm">
      <area shape="rect" coords="290,172,333,250" alt="Phone" href="phone.htm">
    </map>
  </body>
</html>
```

HTML Audio/Video

- Before HTML5, audio/video files could only be played in a browser with a plug-in (like flash).
- Audio and Video can contain one or many <source> tags with different sources. The browser will choose the first source it supports.
- The **controls** attribute adds audio controls, like play, pause, and volume.

HTML Markup

```
<video width="320" height="240" controls>
  <source src="movie.mp4" type="video/mp4">
  <source src="movie.ogg" type="video/ogg">
  Your browser does not support the video tag.
</video>

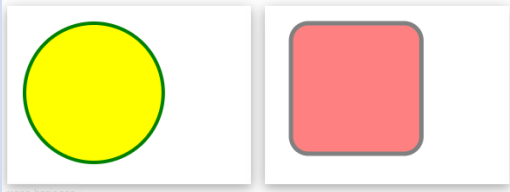
<audio controls>
  <source src="song1.mp3" type="audio/mpeg">
</audio>
```

HTML Entities & Special Characters

- Reserved characters in HTML must be replaced with character entities.
- If you use the less than (<) or greater than (>) signs in your text, the browser might mix them with tags.
- Therefore, character entities should be represented with ***&entity_name*** OR ***&#entity_number***.

Result	Description	Entity Name	Entity Number
	non-breaking space	 	
<	less than	<	<
>	greater than	>	>
&	ampersand	&	&
"	double quotation mark	"	"
'	single quotation mark (apostrophe)	'	'
¢	cent	¢	¢
£	pound	£	£
¥	yen	¥	¥
€	euro	€	€
©	copyright	©	©
®	registered trademark	®	®

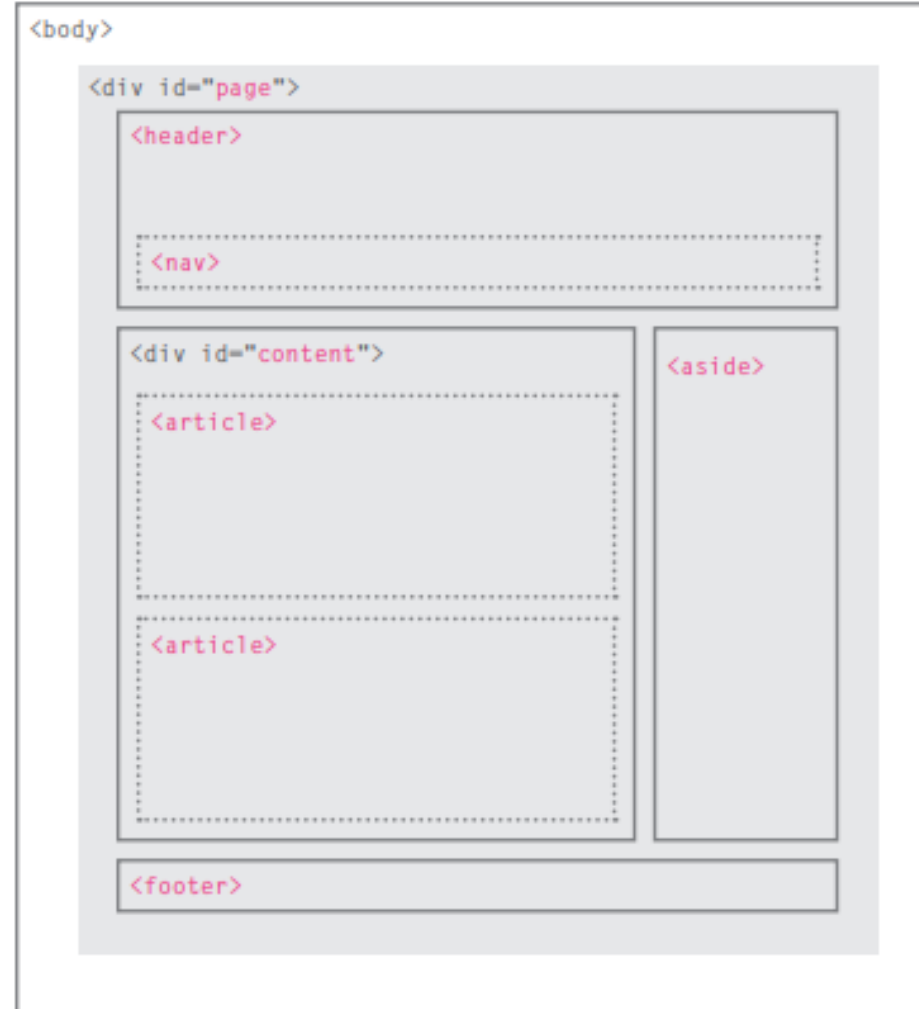
HTML Graphics

HTML Graphics	Example	Purpose
SVG		- The HTML <svg> element allows vector based graphics in HTML - SVG is used to define vector-based graphics for the Web.
Canvas		- The HTML <canvas> element can be used to draw graphics on a web page: - The <canvas> element is only a container for graphics. You must use a JavaScript to actually draw the graphics.

HTML5 Layout Tags

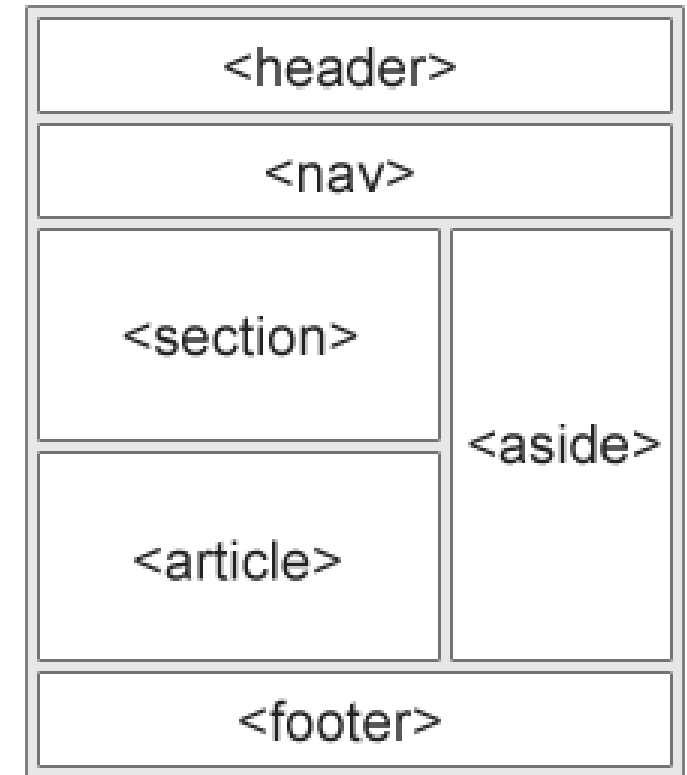
- HTML5 has introduced a new set of elements that help define the structure of a page.
 - <article>
 - <aside>
 - <details>
 - <figcaption>
 - <figure>
 - <footer>
 - <header>
 - <main>
 - <mark>
 - <nav>
 - <section>
 - <summary>
 - <time>

HTML5 Page Structure



HTML5 Semantic Tags

- A semantic element clearly defined its content and meaning to both the browser and the developer.
- Non-semantic elements: Eg:<div>,
- Semantic elements: Eg:<form>, <table>, and <article>



Troubleshooting

- The websites should be thoroughly tested to avoid problems such as:
- **Typos**
 - HTML must be exact in order to work correctly.
 - A single mistyped or missing character can wreak havoc in your webpages.
- **Broken Links**
 - If a user clicks on a link and it doesn't take them where you intend, it is considered a "broken link".
- **Missing Images**
 - Check that the path and file name of the image is exactly correct, including the file extension (.jpg, .gif, etc.)

Exercise

Write the necessary HTML to make your page look just like this one:

This is your first HTML page!

Seems like things are going pretty well so far

Let's add a recipe for some delicious chocolate chip cookies

We're going to borrow a nice picture from General Mills



Here is what the recipe calls for

Ingredients (probably not right...)

- 6 Eggs
- 2 Cups of flour
- 2 Tablespoons of vanilla extract
- 2 Teaspoons of baking powder
- 1 Cup of sugar

Instructions (may or may not be correct)

1. Add eggs, flour and vanilla extract and mix
2. Combine with sugar and baking powder and stir
3. Eat half the cookie batter because it is delicious
4. Take the rest of the cookie batter and make cookies
5. Bake at 350 for 25 minutes
6. Enjoy!

Reading

- Duckett, J. (2011), *HTML & CSS, Design and Build Websites*, Wiley.
- Castro & Hyslop, B.(2011), *HTML5&CSS: Visual QuickStart Guide*, Peachpit Press.
- Robinson, J . (2018), *Learning Web Design, A Beginner's Guide to HTML,CSS, JavaScript, And Web Graphics*, O'Reilly.
- Online :
<http://www.w3schools.com/html/>